Minsu Kim, Ph.D.

Postdoctoral Candidate

Email. kmmns0613@gmail.com
Linkedin. https://www.linkedin.com/in/minsukim0613
Webpage. https://kmmns0613.github.io/minsu-kim.github.io/CV/

EDUCATION

Mar. 2019 – Feb. 2025	**Konkuk University** Department of Bioscience & Biotechnology *Integrated M.S. and Ph.D.* Program (Advisor: Dr. Yoon Kyung Choi)	Seoul, South Korea
Mar. 2013 – Feb. 2019	**Konkuk University** Department of Biomedical Science & Engineering	Seoul, South Korea

RESEARCH EXPERIENCES

1. **Independent Researcher** | Space Neurobiology • Transcriptomics • Ferroptosis (May 2025 – Present)
   - Conducted integrative transcriptomic analyses of region-specific gene expression changes in mouse brain after 35-day spaceflight using NASA GeneLab RNA-seq data.
   - Focused on oxidative stress, iron metabolism, and ferroptosis-related pathways associated with spaceflight-induced brain vulnerability, with interpretations grounded in astrocyte biology and mitochondrial stress mechanisms.
   - Developed OmicsForge, a modular transcriptomics framework for NASA GeneLab datasets, improving reproducibility and interpretability.

2. **Ph.D. Candidate** | Traumatic brain Injury • Inflammation • Mitochondria (Mar. 2019 – Apr. 2025)
   - Investigated immune cell–mediated neuroinflammatory mechanisms following traumatic brain injury, with a focus on peripheral immune cell infiltration and secondary injury processes.
   - Elucidated astrocyte mitochondrial dysfunction in TBI models and its impact on neuronal differentiation and neuroprotective responses.

RESEARCH INTEREST

- Space Neurobiology
- Brain Responses to Microgravity and Cosmic Radiation
- Mitochondrial, Oxidative, and Cellular Stress under Extreme Environments
- Astrocyte- and Immune Cell-mediated Mechanisms of Neurodegeneration
- Translational Approaches for Spaceflight Countermeasures

EXPERIMENTAL NEUROBIOLOGY SKILLS

- **In vivo mouse models of brain injury and stress** (TBI modeling, drug administration)
- **Cell culture** (Macrophages, Astrocytes, Neural Stem Cells, Endothelial Cells, Pericytes)
- **Western blotting**
- **Immunofluorescence-based analysis** (Fresh-frozen Brain/Retinal sections; Cultured cells)
- **Quantitative image analysis** (ImageJ)

COMPUTATIONAL & TRANSCRIPTOMIC SKILLS

- **R** (Transcriptomic data analysis, Gene set enrichment analysis and Data visualization)
- **Python (**Data visualization & Statistical analysis)

PUBLICATIONS (SCI ONLY)

1. Hyungsu Kim, Sunhong Moon, **Minsu Kim**, Hyungkeun Oh, Jinhong Park, Suji Kim, Taehyung Yoo, Ji-Yoon Kim, Yonghee Kim, Young-Myeong Kim, Yoon Kyung Choi, "Upregulation of astrocytic mitochondrial functions via Korean red ginseng-induced CREB-BK**α**-HIF-1**α** axis through L-type Ca^{2+} channel subunits **α**1C and **β**4". *J Cereb Blood Flow & Metabolism,* May 2:271678X251332760 (2025).

2. Sunhong Moon, Jinseo Park, Sueun Kim, **Minsu Kim**, Hui Su Jeon, Hyungsu Kim, Young-Myeong Kim, ji-yoon Kim, Yoon Kyung Choi, "Korean Red Ginseng-induced astrocytic HIF-1α: A key regulator of neuroglobin derived from neural stem cell differentiation in physiologic retina and brain". *Journal of Ginseng Research,* Mar;49(2):189-196 (2025).

3. Eunyoung Jung, Ye Eun Kim, Hui Su Jeon, Myeongjong Yoo, **Minsu Kim**, Young-Myeong Kim, Seong-Ho Koh, Yoon Kyung Choi, "Chronic hypoxia of endothelial cells boosts HIF-1α-NLRP1 circuit in Alzheimer's disease". *Free Radical Biology and Medicine*, Aug 1;204:385-393 (2023).

4. Hui Su Jeon[†], Chang-Hee Kim[†], **Minsu Kim**, Sunhong Moon, Yoon Kyung Choi, "Korean red ginseng mediates mitochondrial membrane potential repair via the Tom22-Tom20-SIRT2 pathway in astrocytes". *Conditioning Medicine* Oct, 5(3), 105-111 (2022).

5. Sunhong Moon, Chang-Hee Kim, Jinhong Park, **Minsu Kim**, Hui Su Jeon, Young-Myeong Kim, Yoon Kyung Choi, "Induction of BVR-A Expression by Korean Red Ginseng in Murine Hippocampal Astrocytes: Role of Bilirubin in Mitochondrial Function via the LKB1-SIRT1-ERRα Axis". *Antioxidants (Basel)*, Sep 1;11(9):1742 (2022).

6. Jinhong Park, Minjae Lee, **Minsu Kim**, Sunhong Moon, Seunghee Kim, Sueun Kim, Seong-Ho Koh, Young-Myeong Kim, and Yoon Kyung Choi, "Prophylactic role of Korean Red Ginseng in astrocytic mitochondrial biogenesis through HIF-1α". *Journal of Ginseng Research*, May;46(3):408-417 (2022).

7. **Minsu Kim**, Sunhong Moon, Hui Su Jeon, Sueun Kim, Seong-Ho Koh, Mi-Sook Chang, Young-Myeong Kim, Yoon Kyung Choi, "Dual Effects of Korean Red Ginseng on Astrocytes and Neural Stem Cells in Traumatic Brain Injury: The HO-1-Tom20 Axis as a Putative Target for Mitochondrial Function". *Cells* Mar 4;11(5):892 (2022).

8. **Minsu Kim**, Joohwan Kim, Sunhong Moon, Bo Young Choi, Sueun Kim, Hui Su Jeon, Sang Won Suh, Young-Myeong Kim, Yoon Kyung Choi, "Korean Red Ginseng Improves Astrocytic Mitochondrial Function by Upregulating HO-1-Mediated AMPKα-PGC-1α-ERRα Circuit after Traumatic Brain Injury". *International Journal of Molecular Sciences*, Dec 3;22(23):13081 (2021).

9. **Minsu Kim**, Hyejung Mok, Woon-Seok Yeo, Joong-Hoon Ahn, and Yoon Kyung Choi, "Role of ginseng in the neurovascular unit of neuroinflammatory diseases focused on the blood-brain barrier". *Journal of Ginseng Research*, Sep;45(5):599-609 (2021).

MANUSCRIPTS (IN PREPARATION / UNDER REVIEW)

1. **Minsu Kim**, Wonjin Park, Tae-Kyeong Lee, Joohwan Kim, Joon Ha Park, Moo-Ho Won, Young-Guen Kwon, Ji-Yoon Kim, Ken Arai, Eng H Lo, Young-Myeong Kim, and Yoon Kyung Choi, "Defending function of inducible heat shock protein 70 against vascular permeability". Manuscript under review.

2. **Minsu Kim**, Issac Kremsky, Michael Pecaut, and Xiao Wen Mao, "A Ferroptosis-Related Transcriptional Signature in the Brain After a 35-day Spaceflight". Manuscript under review.

3. Thapa Magar, Til Bahadur, **Minsu Kim**, Yoonha Kim, Eun Young Choi, Yoon Kyung Choi, Dong-Eun Kim, Kyung Bo Kim, Ji Eun Park, "Pharmacokinetics, target engagement of an immunoproteasome subunit LMP2 inhibitor AR-01, and its anti-Alzheimer's effects in rodents". Manuscript under review.

INTERNATIONAL CONFERENCE

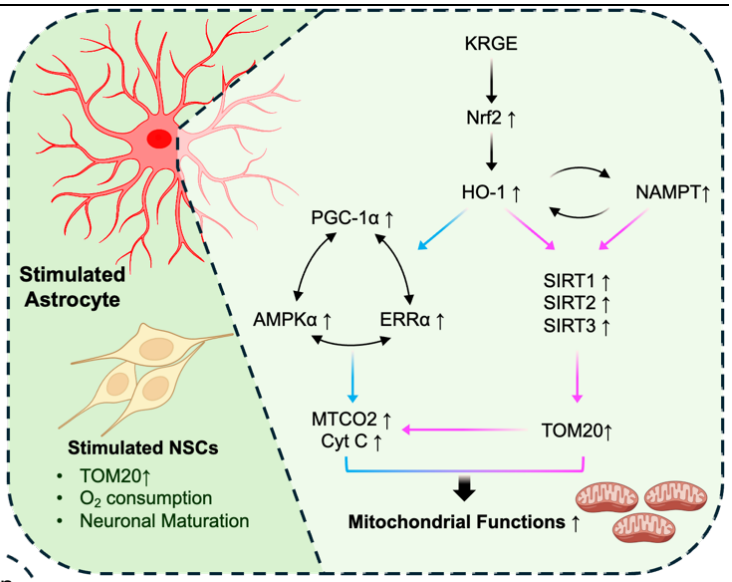
1. **Minsu Kim**, "Astrocyte Ferroptosis in the Brain After Long-Term Spaceflight: Insights from NASA GeneLab RNA-seq Data". *American Society of Gravitational and Space Research.* Dec 3-6, 2025, Phoenix, AZ, USA. (Oral Presentation)

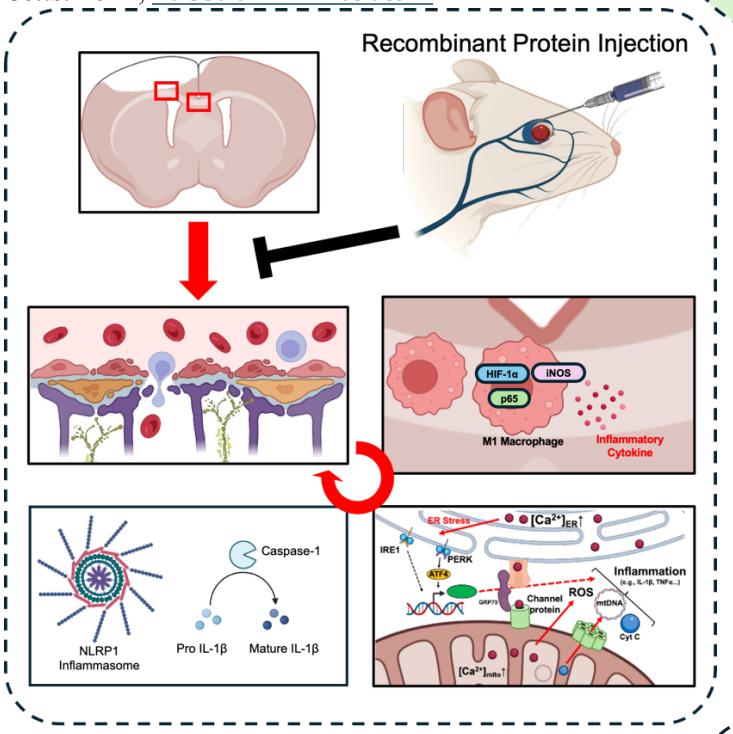
RESEARCH SUMMARY

Two HO-1–Dependent Pathways That Enhance Mitochondrial Function in Damaged Astrocytes

Intake of Korean Red Ginseng Extract increases the expression of Heme Oxygenase 1 of astrocytes in the brain of TBI mouse model. HO-1 then increases the circuit of three proteins (AMPKα-PGC-1α-ERRα) or increases SIRT protein levels by interacting with NAMPT. Increased TOM20 expression by SIRTs and the circuit induces mitochondrial protein (MTCO2, Cytochrome C). These two independent pathways are dependent on HO-1 activity, as demonstrated by treatment with SnPP (HO-1 activity inhibitor).

Int J Mol Sci. 2021, 10.3390/ijms222313081.
Cells. 2022, 10.3390/cells11050892.



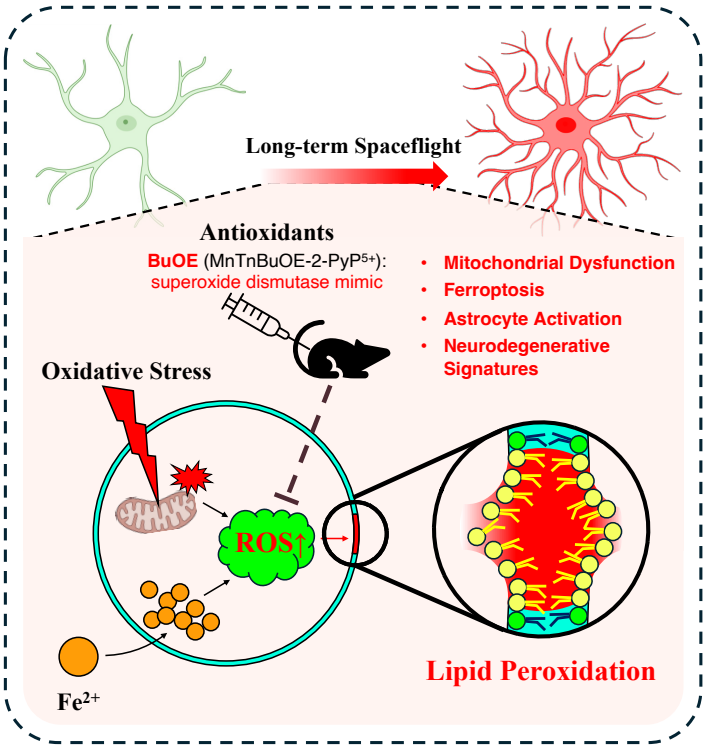


Protecting Blood-Brain Barrier Integrity Limits Immune Cell Infiltration and Inflammation After Traumatic Brain Injury.

After traumatic brain injury (TBI), immune cells that recognize damage signals infiltrate brain tissue through the blood-brain barrier (BBB), become activated and polarized, and acquire pro-inflammatory functions. These cells spread along white matter regions and are exposed to hypoxia, releasing cytokines that amplify inflammation in neighboring cells. Stressed mitochondria contribute to this process by upregulating inflammatory proteins and mitochondrial channel proteins. This study aims to characterize how a specific recombinant protein, delivered via retro-orbital injection after TBI, preserves BBB integrity and limits immune cell infiltration by suppressing inflammation at the molecular level.

2026, Draft Manuscript in Preparation.

Astrocyte Ferroptosis in the Brain After 35-Day Spaceflight: Insights from NASA GeneLab RNA-seq Data

Using transcriptomic datasets from NASA GeneLab (OSD-682, 685, 698, 699), which include spatial transcriptomics after 35 days aboard the ISS, I investigated region-specific molecular responses to long-duration spaceflight. Preranked fast GSEA and single-sample GSEA revealed a strong ferroptosis-related signature in the hippocampus, particularly within the dentate gyrus, driven by elevated intracellular iron, oxidative stress, and OXPHOS-associated ROS production. These stress-response pathways also correlated with Alzheimer's disease–related pathways, suggesting that ferroptosis may contribute to neurodegenerative signaling during spaceflight. Integrating these findings with prior imaging data, astrocytes emerged as a plausible cellular source of this ferroptosis-like response.

2026, Draft Manuscript under Review.